

Family-centered measures of access to early care and education

Elizabeth E. Davis^{a,*}, Won F. Lee^b, Aaron Sojourner^c

^a Department of Applied Economics, University of Minnesota, 316B Ruttan Hall, 1994 Buford Avenue, St Paul, MN 55108–6040, United States

^b Department of Applied Economics, University of Minnesota, 253 Ruttan Hall, 1994 Buford Avenue, St Paul, MN 55108–6040, United States

^c Department of Work and Organizations, Carlson School of Management, University of Minnesota, 321 19th Ave South, 3–300Y, Minneapolis, MN 55455–0438, United States

ARTICLE INFO

Article history:

Received 5 June 2017

Received in revised form 22 July 2018

Accepted 5 August 2018

Available online 14 September 2018

Keywords:

Child care

Early education

Child care access

Two-stage floating catchment area

Slots per tot

ABSTRACT

This study proposes new family-centered measures of access to early care and education (ECE) services with respect to quantity, cost, and quality and uses them to assess disparities in access across locations and socio-demographic groups in Minnesota. These measures are distance-based and use available geographic data to account for the fact that families can cross arbitrary administrative boundaries, such as census tract or ZIP code lines, and thus better reflect the real experiences of families than conventional area-based measures. Combining synthetic family locations simulated from Census demographic and geographic data and information on ECE provider locations, we calculate travel time between the locations of families with young children and ECE providers to measure families' access to providers of different types. The results yield a map of areas with low and high relative ECE access. The average family in Minnesota lives in a location where there are nearly two children for every nearby slot of licensed capacity, however, access to ECE supply varies considerably at the local level. The supply measure can also serve as a weight useful in computing family-centered measures of ECE quality and access costs, incorporating both prices and travel costs, to further characterize the local ECE market from the perspective of families. Improving the measures of variation in families' access to ECE quantity, cost, and quality is valuable as policymakers consider expansions to public supports for early learning and ECE entrepreneurs decide where to invest.

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1. Introduction

Access to affordable, high-quality early care and education (ECE) is important because of the dual role of ECE in facilitating employment of parents and in supporting child development. As our understanding of the importance of early childhood learning environments expands, disparities in access to high-quality ECE have become of increasing concern to policymakers. The reauthorization of the Child Care and Development Block Grant (CCDBG) in 2014 established new requirements for states to improve access to high quality ECE (US DHHS, 2014). Studies that examine community-level measures of the availability of ECE conclude that disparities in access are related to the sociodemographic and economic characteristics of local communities (Bassok, Fitzpatrick, & Loeb, 2011; Bassok & Galdo, 2016; Fischer, Nelson, Mikelbank, & Coulton, 2008;

Fuller & Liang, 1996; Gordon & Chase-Lansdale, 2001; Loeb, Fuller, Kagan, & Carrol, 2004).

To ensure adequate access to early care and education, particularly for low-income families, policymakers and planners need accurate measures of access in order to target resources to locations with greatest need. After a comprehensive review of the literature, Friesen, Lin, Forry, and Tout (2017) write, “Recent federal policy changes have placed new requirements on states to demonstrate and document their efforts to improve access to high-quality early care and education (ECE), and have made clear the urgent need for a shared understanding of this concept.” Many factors influence families' access to ECE services. Past studies of access have mostly relied on area-based measures of the availability of ECE options. They typically partition a region into non-overlapping zones, using ZIP code, county, census tract, or municipal boundaries for example. It is common to compare a measure of supply, such as the number of providers, provider capacity, or a count of child care workers, to the number of children within each area (Bassok et al., 2011; Casper & O'Connell, 1998; Cochi Ficano, 2006; Fischer et al., 2008; Fuller & Liang, 1996; Gordon & Chase-Lansdale, 2001; Jacobson, 2001; Neidell & Waldfogel, 2009; Yamauchi, 2010). By defining relevant

* Corresponding author.

E-mail addresses: edavis@umn.edu (E.E. Davis), leex5089@umn.edu (W.F. Lee), asojourner@umn.edu (A. Sojourner).

areas based on administrative boundaries, these conventional measures ignore families' ability to travel across boundaries to find care. They implicitly consider all supply in the same area as a family as equally accessible to that family and all supply outside the area as completely inaccessible. In addition, standard area-based measures treat all families in the area as equally likely to demand all in-area supply and all families outside the area as irrelevant, when in reality, families will cross area administrative boundaries to find care.

This paper builds on the prior literature by developing a new set of family-centered, distance-based measures of access to ECE services' quantity, cost, and quality. We measure the quantity of supply provided near where families with young children live and the number of other young children living nearby those providers potentially competing for the supply, the travel time required to reach that supply, and the cost and quality of the supply. We use Census data on the population of Minnesota families with young children combined with data on all the state's licensed center and family child care providers, public preschool, and Head Start providers. Our distance-based measures do not implicitly restrict families from crossing administrative boundaries in accessing nearby ECE services. Our supply (quantity) measure uses an enhanced two-stage floating catchment area design (Luo & Qi, 2009), which formalizes the idea that greater access for a family means having more supply closer by with fewer other children potentially competing for that supply. It captures variation in the nearby supply of ECE services from families' perspectives adjusting for the child population nearby each provider, where "nearby" is determined by drive time.

We extend the logic of this approach to propose new measures of families' cost of access to ECE services and the quality of ECE services accessible to families. The cost-based measures reflect both the monetary price of care and the cost of travel between the family's location and each nearby provider. The quality measure is basically a family-specific average of local provider quality. Each of these dimensions of access – quantity, cost, and quality – is different and important. Together, they give a holistic, textured view of ECE services from families' perspectives. While additional factors influence families' selection of ECE providers, such as the hours of care (for example, whether evening or weekend hours are needed and offered), the cultural and language fit, and even whether the family has information about the provider, these are beyond the scope of this study (Carrillo, Harknett, Logan, Luhr, & Schneider, 2017; Davis & Connelly, 2005; Han, 2004; Kim & Fram, 2009; Liang, Fuller, & Singer, 2000; Peyton, Jacobs, O'Brien, & Roy, 2001; Radey & Brewster, 2007).

These new methods measure access to ECE services at a particular location, characterize it across subpopulations of families with young children, and enable examination of disparities in access for different groups of families defined by combinations of geography, racial and ethnic background, and income level. We demonstrate this approach using data from Minnesota in 2015 to estimate these measures among different communities in the state and make comparisons across different kinds of families.

2. Literature review

A number of studies have shown how child care supply varies across communities in ways associated with community characteristics. Notably, child care availability is lower in rural areas and in lower-income communities compared to urban and higher-income areas (Cochi Ficano, 2006; Gordon & Chase-Lansdale, 2001; Hofferth & Wissoker, 1992). Other studies have found disparities across communities in the availability of preschool options (Bassok

& Galdo, 2016), particularly high-quality settings (Bassok et al., 2011).

While there is widespread acknowledgement of the local nature of child care markets (Queralt & Witte, 1998), operationalizing what is meant by *local* has been a challenge. Chipty and Witte (1998), for example, use distance between child care centers (based on five-digit ZIP code) to define child care markets. Based on a nationally-representative sample, the National Survey of Early Care and Education estimated that, on average, families used center-based arrangements within five miles of home for children under age 6 (National Survey of Early Care and Education Project Team, 2016). A number of recent studies have mapped information on the geographic locations of child care providers, often to provide information to parents about local care options (Alexander et al., 2016; Hardy et al., 2016). For example, a recent report provides a map of school-based and licensed center-based and family child care programs for two ZIP codes in Minneapolis (Overson & McConnell, 2015). A related study looks at inequality of access to child care using a "spatial mismatch" approach, analyzing the distribution of child care centers to that of families with young children by ZIP code, which fundamentally is an area-based measure (Covington, 2007).

A number of studies have focused on access to high-quality settings and the role of child care subsidies or public funding in facilitating access. Hatfield, Lower, Cassidy, and Faldowski (2015) examined differences in access to high-quality ECE across communities in North Carolina by estimating the relationship between the quality of ECE programs and community characteristics and funding sources. The study found that higher quality centers (those with higher scores in the quality rating system) were more likely to be located in more affluent communities and generally received more funding from Head Start and public preK funds. Johnson, Ryan, and Brooks-Gunn (2012) highlighted the importance of cost to families in determining access and the variation in quality across private and publically funded programs. They found that families receiving child care subsidies used higher-quality settings than similar families who did not use publically-funded ECE, yet the subsidized families also used lower-quality care compared to those in public programs such as Head Start. Marshall, Robeson, Tracy, Frye, and Roberts (2013) found that families in Massachusetts who received child care subsidies reported higher-quality care than similar families on the subsidy waiting list, and were more satisfied with care. While there is no single measure or definition of access in the literature, these studies highlight the importance of cost and quality to families as important aspects of measuring access.

Despite the popularity of area-based measures in the child care literature, the key limitation of these measures stems from the fact that the unit of analysis is based on administrative or political boundaries that are arbitrarily defined from the perspective of families deciding on child care for their children. McLafferty (2003) provided an analogous discussion of area- and distance-based measure of access in the health care literature. Area-based measures of access to ECE supply typically measure access by area using the area's ratio of the total number of slots divided by the total number of children or ratio of children to child care workers (Bassok et al., 2011; Fuller & Liang, 1996; Gordon & Chase-Lansdale, 2001; Neidell & Waldfogel, 2009). The potential for information loss and statistical bias when using area-based measures is known as a modifiable areal unit problem (MAUP) and has been a longstanding concern in geography (Gehlke & Biehl, 1934; Openshaw, 1984).

2.1. This study

In contrast to much of the prior literature's reliance on area-based measures of access, this paper develops distance-based measures. However, distance-based measures presume knowledge

of both ECE provider locations, which are commonly known, and family residential locations, which are generally geographically-bounded but not known exactly. To protect privacy, public data on families are available only by area, for example, counts by families by type within a Census block, block group, or tract. We approximate the distribution of family locations by combining that information with an assumption that families are uniformly distributed within a Census block (outside known, non-residential areas such as lakes, rivers and state parks). This approach harnesses all available data on Census blocks' shapes and locations and the distribution of family types across Census spatial units, along with data on the road network and nonresidential areas.

We demonstrate that our distance-based measures of access provide a more accurate description of the set of provider choices near to the family, as some of the providers may be located near a family but across an administrative boundary. While we estimate that only 16% of households would have to cross a county line to get to the closest higher-rated provider in our data, 71% of households would cross census-tract boundaries and 87% would cross a census-block-group boundary line. Measures of access to child care based on administrative boundaries miss many options relevant to families. To illustrate this issue, our analysis will compare results obtained with the same data under distance-based and area-based measures. The online Supplementary material presents a simplified example comparing the two approaches.

3. Data, measures and methods

3.1. Data on child care providers

Data on ECE providers in Minnesota in 2015 were obtained from the NACCRRAware data system managed by Child Care Aware of Minnesota (a child care resource and referral network). Information on each provider in the dataset includes total licensed capacity, enrollment by age group, prices by age group, and street address. The dataset identifies four types of providers: (i) child care centers (CCC), (ii) licensed family child care (FCC), (iii) Head Start and Early Head Start programs, and (iv) School Readiness pre-kindergarten programs run by school districts. We group (iii) and (iv) into a public provider category. Public providers that operate more than one program at the same spatial location were counted as one provider. Private providers (CCC and FCC) are those not identified in the data as Head Start, Early Head Start or school-based programs.

In order to create access measures based on quantity and cost, we need information about each provider's capacity and prices in addition to its location. For private providers, the quantity measure for children age 0–4 is based on the provider's total licensed capacity minus current school-age enrollment. While providers may prefer to enroll fewer than their licensed maximum, this number represents the highest possible number of young children that can be served onsite at a time.¹ For public providers – both Head Start and school-based pre-kindergarten and preschool programs, information on location is provided but data on capacity are less available. One quarter of the Head Start programs are missing capacity information, so we use the median capacity in Head Start programs by region to impute capacity to these programs. For school-based programs, very few report capacity information. The few who do so report capacity of about 20 children per program. We use this number, approximating one classroom, to impute capacity for each school-based program. This data limitation war-

rants caution in interpreting the access measures for public ECE providers.

For the purposes of this study, provider quality is measured by ratings in Minnesota's Quality Rating and Improvement System (QRIS), named Parent Aware. The star level that a program receives is based on points earned in four domains: physical health and well-being; teaching and relationships; assessment of child progress; and teacher training and education (Cleveland, Li, Orfali, & Tout, 2016). Programs that choose to be fully rated receive financial and coaching supports. An initial evaluation of the Parent Aware system concluded that the star-rating system identified important differences in quality practices across programs (Tout et al., 2016). Whether these criteria are in accord with parents' perceptions of quality is an important issue but is beyond the scope of this study (Cryer, Tietze, & Wessels, 2002).² Information on Parent Aware ratings (one to four stars) for those providers who voluntarily participate is included in the dataset. Providers who did not participate in Parent Aware as of 2015 have no rating information. While we acknowledge that these providers may be of any quality level, we group them together into an "unrated" category.

To calculate a weekly access cost measure, we aim to measure the price for weekly, full-time care at each provider. Private (or market-based) providers report prices by age group: infant, toddler, preschool age and school age. To distill this information into a single price for each provider for the care of young children, we use weights based on the number of months a child age 0–4 would spend in each age group based on Minnesota regulations. For example, in a family child care setting, infants are children 12 months and younger, toddlers are aged 12–24 months, and preschool age is 24–60 months. Thus, for each family care provider, we calculate the weighted average of its infant, toddler, and preschool weekly prices with weights 12/60, 12/60, and 36/60, respectively. A similar process is used for child care centers with slightly different weights because the regulatory age-group definitions differ in Minnesota. Substantive results of the analysis are similar if we focus on one price, such as the price for preschool-age children, instead.

Forty percent of private providers do not report prices on a weekly basis, but instead report hourly or daily prices. These prices were converted to a weekly basis using the ratio between weekly and daily prices or between weekly and hourly prices in the market among providers that report prices in both time units. For private providers missing all price information (27.8%), prices were imputed by scoring their observed characteristics using a spatially-weighted regression of price controlling for type of care (center or family child care), location, Parent Aware quality rating (if rated), and provider's years of experience and estimated using the providers with both prices and characteristics observed. Additional details on conversion of prices to weekly rates and imputation of missing price information can be found in the online Supplementary material. Price information was available for March, June and September 2015. We average over the three months for each provider to obtain a 2015 price.³ The data do not contain price information for public providers. We assume they have a zero money price for parents. Any payments by parents for public programs are not observed but would be easy to integrate into the analysis if they were.

² There is an extensive literature that examines how to define, measure and improve child care quality and that investigates the linkages between quality and child outcomes (see, for example, Cassidy et al., 2005; Peisner-Feinberg et al., 2001).

³ For most providers, the prices are updated once per year in the database. Given the small differences in prices across the three quarters, it made little difference whether we averaged or used the most recent value.

¹ Note: Most prior studies on access use total licensed capacity and do not state whether they adjust for school-age children, except to the extent that they do not include school-age only programs.

Table 1
Characteristics of ECE providers in Minnesota, 2015.

	Mean	Std. dev.
Family child care (share)	0.842	0.364
Child care centers (private) (share)	0.099	0.299
Public (Head Start or public preK) (share)	0.058	0.234
Weekly price – infant	152.9	53.76
Weekly price – toddler	143.9	45.60
Weekly price – preschool age	135.8	41.90
Weekly price – weighted age groups	143.1	45.04
Capacity – family child care	9.33	1.65
Capacity – child care centers (private)	66.47	39.43
Capacity – public (Head Start or public preK)	28.25	41.58
Unrated and lower rated providers (share)	0.887	0.317
Highly rated providers (private) (share)	0.058	0.233
Minneapolis-St Paul MSA (share)	0.468	0.499
Number of observations (N)	11,989	

Note: Provider rating is the provider's current star level in Parent Aware, if they are rated. Public providers are assumed to have a price of zero for parents (which are treated as missing in calculating the mean price in this table). The Minneapolis-St Paul metropolitan statistical area (MSA) is the largest in Minnesota and consists of 11 contiguous counties. The remaining counties in Minnesota are collectively referred to as "Greater Minnesota" and include more rural parts of the state as well as smaller metropolitan areas.

3.1.1. Characteristics of the ECE providers in Minnesota in 2015

There are about 12,000 ECE providers in the dataset, covering nearly all of the licensed providers in Minnesota in 2015, though new providers may be delayed in entering the dataset and systematic data are not available on friend, family and neighbor care. As shown in Table 1, only 10% of all licensed providers are private child care centers, with an average total licensed capacity of 66 slots. Nearly 6% of providers are Head Start and public pre-kindergarten programs, which we group together as "public" programs, with an average capacity of 28 slots. The majority of providers (84.2%) are family child care providers with an average capacity of 9.3 slots for children age 0–4. Among private providers, both centers and family based, weekly prices ranged from an average of \$153 for infants to \$136 on average for preschool age children. To examine regional differences, we compare the Minneapolis-St Paul metropolitan statistical area (MSA) and "Greater Minnesota," which includes more rural parts of the state as well as smaller metropolitan areas. Nearly half of the providers (47%) are located in the 11-county metropolitan area around the cities of Minneapolis and St. Paul.

We divide providers into three groups: (1) highly rated market providers, which includes providers who have a three- or four-star rating in Parent Aware and are not a public provider; (2) public programs, such as Head Start, Early Head Start, and School Readiness pre-kindergarten programs which are designated as highly rated (four stars) in the Parent Aware system and also do not report a price charged to parents; and (3) all other providers, which includes both unrated providers, i.e., those who do not have a rating in the voluntary Parent Aware QRIS and lower-rated programs that have a one- or two star rating in Parent Aware. This three-way classification allows us to focus on the two important dimensions: first whether the provider known to be highly rated, and second, whether they charge parents or are a public program. A large majority (83.4%) of providers were not rated in the Parent Aware QRIS in 2015 as the voluntary system has been rolling out over time (Cleveland et al., 2016). While unrated providers can be of any quality level, as the QRIS expands over time, future analysis will be able to examine quality differences more fully.

3.2. Data on families and family locations

To construct distance-based measures of families' access to child care, the ideal dataset would have exact residential locations of all families with young children as well as information on their socioeconomic characteristics. Exact locations, and information on the income and race of actual families with children under age 5 are not available due to privacy concerns. We approximate the spatial distribution of families' residential locations and their demographic characteristics by combining information from the Census American Community Survey (ACS) 2011–2015 five-year estimates (2015 ACS) and the 2010 decennial Census (Manson, Schroeder, Van Riper, & Ruggles, 2017).

In particular, we estimate the joint distribution of race, income, and number of children under 5 for families in each Census block combining data from the 2010 decennial Census and 2015 ACS using assumptions and methods detailed in a technical Appendix in the online Supplemental material, and described briefly here. From the 2015 ACS, we obtain recent estimates of the number of families with any child under age 5 by census block group. To get more exact geographic information about where families with young children live, we incorporate block-level data from the 2010 decennial census. Census blocks cover much smaller geographic areas than Census block groups or tracts (there are over 250,000 census blocks in Minnesota and 1335 census tracts). The likelihood of a synthetic family location being in a particular Census block is proportional to the estimated likelihood that a real family with children under age 5 lives in that block. The exact location of any synthetic family within a block is random assuming a uniform distribution of families within the block boundaries.⁴ We restrict the possible locations for synthetic families to exclude lakes, rivers and state parks.

To model the distribution of families within block groups accounting for racial clustering, we assume that, conditional on race, the distribution of families with young children across blocks within block groups is the same in 2015 as in 2010. In essence, we apply the race-income joint distribution among families with a young child at the lowest-available geographic level to any sub-units in order to estimate the joint distribution of race, income and location of the synthetic sample. We use 2015 ACS census tract-level estimates of the joint distribution of race and income among families with young children to approximate the race and income of families at the block level.

The ACS estimates that 259,503 families with children under age 5 lived in Minnesota in 2015. With about 12,000 ECE providers, the number of provider-family location pairs is extremely large, approximately 3 billion pairs. For computational ease, we draw a set of synthetic family locations numbering one quarter of the total number of Minnesota families ($N=64,875$ synthetic family locations), so each location represents four families. The 64,875 family locations are randomly drawn based on the estimated number of families with young children living in each Census block. For each family, we impute their number of children under 5 equal to their block group's average number of young children per family with any young children. At the end, each synthetic family location has an associated number of young children equal to four times the block-level average and a joint probability distribution over race and income level.

Travel time between each synthetic-family location and any ECE provider is estimated using *osrmtime*, a Stata command that calculates driving time between two geocoded points (Huber & Rust,

⁴ We use Geospatial Modelling Environment (GME) software for assigning family locations.

2016).⁵ When computing travel time, the algorithm locates the nearest point on the road network and travel times are computed from that point. This functionally excludes locations off the road network and ensures the assignment of synthetic families to random habitable locations within a block. For each synthetic family location, the travel time to each provider within 20 min driving time is calculated. For 477 families of 64,875 total, there are no providers within 20 min driving time. For these families, we search farther, find the closest Minnesota provider of each of our three types, and include these.

3.3. Access measures

3.3.1. Quantity measure: slots adjusted for nearby population

We create a measure of nearby supply adjusting for the number of children nearby each provider using the enhanced two-stage floating catchment area (E2SFCA) method pioneered in geography (Luo & Qi, 2009). Our measure is conceptually similar to the commonly-used measure of the quantity of child care informally called the number of “slots per tot” – the total capacity of providers (slots) divided by the number of young children – in a particular area. Our measure differs in that it is family-centered and assumes families are interested in nearby providers whether or not they are located in the same (administratively-defined) area. The E2SFCA method measures the supply of slots within a family-specific catchment area and adjusts each nearby provider's capacity based on the number of young children nearby that provider who might compete for the provider's slots. The two stages refer to the two catchment areas: first one for the provider and then another for the family (Luo & Wang, 2003a, 2003b). More technical detail on the creation of the E2SFCA measure can be found in the online supplemental material. Briefly, the E2SFCA measure for each family is computed in two stages:

Stage 1: Calculate the (distance-weighted) capacity-to-population ratio for each provider. For each provider, find all families with young children within the 20-min drive-time catchment area around the provider's location. Summing across families gives the number of young children nearby that provider, which proxies for the strength of potential demand for that provider's slots. However, children closer to a provider are more relevant to provider demand, so we discount farther away children using the Gaussian distance decay function (see technical Appendix for more detail). We divide the number of slots at each provider by the (distance-discounted) child population to obtain a slots-to-(nearby young child) population ratio (or SPR). Two providers who have the same capacity will have different SPRs if one has more young children nearer by. This measure captures the notion that a provider's slots will be less accessible to any nearby family if more children are closer to the provider competing for the slots.

Stage 2: For each family, we measure the quantity of local supply based on the summed capacity of nearby providers adjusting for their nearby young-child population. First we identify all ECE providers within the 20-min catchment area around the family's location (or the nearest providers outside that travel time if none are within it). We then sum the slots-to-population ratios (SPR) for all the providers near the family, discounting each provider's SPR for travel time using the distance-decay function. We refer

to the (distance-discounted) sum of SPRs as the family's measure of *nearby supply adjusted for nearby population* or simply *adjusted supply*. The adjusted supply increases if the family has more slots nearer by and decreases if more young children are nearer by those slots. It is very similar in spirit to the area-based slots-per-tot measure, however, it is family-centered and distance-based. A measure of family's access to adjusted supply from each type of provider can be obtained simply by summing only across providers of that type.

3.3.2. Cost measure: total cost of access

Though quantity-based measures provide important information about access to ECE services, families' access also depends on the cost of ECE services. A family's costs to use a particular provider include both the value of time necessary to travel to the provider and the money price charged. We define the weekly total cost of access at a specific provider for a specific family to be the sum of the weekly travel costs plus the price of weekly full-time care reported by the provider. Weekly travel time is estimated by multiplying the one-way travel time by 10, to capture five round trips per week. We use \$10 per hour to capture the opportunity cost of time. Thus, weekly travel cost is 10 dollars per hour times travel time per trip times 10 trips per week. The weekly money price of full time care is provider-specific and measured as described previously in the section on providers. The total cost of access for a given family for a specific provider is the sum of the travel cost and money price. Combining travel cost and money price into a total cost measure is a transparent way to account for the fact that prices may be higher in urban areas but travel times longer in rural areas. In contrast, conventional area-based measures of cost, such as average price in area, ignore travel costs, and assume all in-area providers are equally relevant to each family while assuming all out-of-area providers are irrelevant.

We summarize the total cost of access in the local market for each family in two ways: first by looking only at the nearest provider, and second, by looking at an adjusted-supply-weighted average of nearby providers. First, one straightforward measure for each family is the cost to access the single provider closest to the family, that is, the provider with the shortest drive time to the family location. Using this measure, the family's access cost will decrease if a new provider opens closer to the family at the same price or if the closest provider's price decreases. The closest-provider cost measure provides a simple summary of cost in the ECE market near the family, but does not account for differences in capacity or child population nearby.

Therefore, we also measure the total access cost (and its components) using a family-specific weighted average of access costs across nearby providers. From the perspective of each family, each nearby provider's (distance-discounted) slots-to-population ratio, SPR, expresses the importance of that provider to the family's adjusted supply. For each family, we compute the average cost of accessing nearby providers by weighting the cost to access each (nearby) provider by the provider's SPR. The difference between two providers' SPRs expresses the difference in the relative importance of their (population-discounted) capacity to that family. Both the closest-provider and the adjusted-supply-weighted average approaches summarize the costs of access in the local market for a family at a particular location. The cost measure based on closest provider is easy to understand but less comprehensive and robust than the weighted average.

3.3.3. Quality measure: highly-rated share of local (adjusted) supply

Given provider-specific measures of quality, the quality of the ECE market around a particular family location can be computed using either of the same two weighting schemes described above for total cost: give all weight to the single closest provider or use

⁵ Note: *osrmtime* takes advantage of two tools: 1. Open Source Routing Machine (OSRM) and 2. OpenStreetMaps. Both are provided by the open source community. Unlike Google Maps which considers real-time traffic obstacles in calculation of travel time, OSRM is a static routing machine and estimated travel time between point A and B can thus be replicated. We compared the travel time estimates of 2000 random location pairs from *osrmtime* to the estimates from Google Maps and found the correlation of the estimates from the two sources was 0.97.

adjusted supply as a provider weight to calculate the average quality of nearby providers. We start with a provider-specific indicator of being highly rated in Parent Aware. For each family, we then measure if the closest provider is highly-rated and measure the adjusted-supply-weighted average of the indicator, yielding a measure of the share of nearby slots that are highly-rated. Minnesota's Parent Aware QRIS was rolled out over time, reaching all counties only in 2015. As a result, the participation rate of providers was relatively low in 2015. Our results demonstrate the concept of building family-specific quality measures but caution is warranted in drawing substantive conclusions from the specific results. The method can apply to any available quality measure, such as a QRIS rating, an indicator of accreditation, an observational rating, a set of indicators of compliance violations, child:adult ratio, or educational credentials of the teachers.

3.4. Analysis methods

The main analytic approach uses a cross-sectional design and descriptive statistical analysis methods to compare the measures by location and demographic subgroup. We calculated means and distributions for each of the measures (adjusted supply, total cost, weekly price, travel time, and highly-rated share). These measures were analyzed overall and for different provider types (child care center, family child care, and public provider or highly-rated private, public, and other providers) and for different locations and race, ethnicity, and income subgroups. We conduct standard difference-in-means statistical tests for differences across subgroups using bootstrapped standard errors.

4. Results

4.1. Quantity: adjusted supply

Table 2 shows the results for adjusted supply. The average family with a young child in Minnesota lives in a location with slightly more than half a slot nearby (0.56) adjusting for the number of other young children nearby each slot. More simply, the average family has access to about half a slot per child. Turning this around, there are nearly two nearby children for every slot of capacity. Although this phrasing sounds similar to that referring to an area-based, slots-per-tot measure, the measure is operationalized very differently.

Overall, there are fewer ECE providers in Greater Minnesota; however, there are also fewer children competing for the slots of those providers. As a result, the adjusted supply measure is significantly higher for families in Greater Minnesota (0.64) than for those in the Minneapolis-St. Paul metro area (0.51). Families in Greater

Minnesota have less access to supply in private child care centers, 0.14 slots per nearby child for families in Greater Minnesota compared to 0.28 for families in the Minneapolis-St. Paul metropolitan area. Thus, the distribution of families' access to adjusted supply across the three categories of ECE providers (centers, family child care and public providers) varies across the state's two regions. A larger share (56%) of adjusted supply is in private child care centers for families in the Minneapolis-St. Paul metro area compared to families in Greater Minnesota (21%). However, families in Greater Minnesota have a higher level of access to adjusted supply from public slots than do families in the Minneapolis-St. Paul (0.10 slots per nearby child compared to 0.03 slots per nearby child) and a higher share of their supply from public slots (16% compared to 6%).

Small differences in levels of adjusted supply are observed across families in different income groups. Compared to families with income less than the federal poverty line (FPL), both middle income (defined as families with income above FPL and less than 185% of FPL) and higher income (defined as families with income above 185% of FPL) have lower overall access. These differences are statistically significant but not large in magnitude (0.59 vs. 0.56 vs. 0.55 adjusted slots per nearby young child). Low-income families have more access to public capacity relative to middle and higher income families (0.081 vs. 0.066 vs. 0.051 adjusted slots). The higher adjusted supply of public capacity near low-income families may reflect intentionally higher public investment in ECE capacity in low-income communities (Fuller & Liang, 1996). Private supply in child care centers and family child care homes is similar across the income categories.

Average adjusted supply is similar across the six racial and ethnicity categories defined in the American Community Survey, and is highest for American Indian families, due to more adjusted supply from public providers (Table 3). Access to private child care center slots is lower for white families and American Indian families compared to those of other groups, in part because these families are disproportionately located in rural parts of the state with fewer centers. Public slots make up about 9–10% of adjusted supply for white (non-Hispanic), African-American and Hispanic families compared to 35% for American Indian families in Minnesota.

This new adjusted measure of supply can be used to analyze variation in smaller geographic areas as well as for different subgroups of the population. The adjusted supply measure is calculated for each synthetic family location and can be plotted on a map to illustrate spatial patterns of supply access. Families living in locations with higher values of adjusted supply have higher ECE access in the sense of geographical proximity to greater supply due to a combination of more slots at nearby providers, providers closer by, and fewer other children closer to those providers. Synthetic family

Table 2
Supply adjusted for nearby children (enhanced two-stage floating catchment area method) in Minnesota, 2015, by region, by income.

	By region			By income group		
	Statewide	Minneapolis – St Paul metro area	Greater Minnesota	Below FPL	Above FPL and less than 185% FPL	Above 185% FPL
Slots adjusted for nearby children						
Any provider	0.558	0.509	0.636***	0.590	0.561***	0.551***
Child care centers (CCC)	0.227	0.284	0.135***	0.234	0.220***	0.226***
Family child care (FCC)	0.274	0.194	0.401***	0.275	0.275	0.273
Public providers	0.058	0.031	0.100***	0.0813	0.066***	0.051***
Share of (adjusted) slots						
Child care centers (CCC)	0.407	0.558	0.212***	0.397	0.392***	0.410***
Family child care (FCC)	0.491	0.381	0.631***	0.466	0.490	0.495
Public providers	0.104	0.061	0.157***	0.138	0.118***	0.093***

Note: The confidence intervals are computed using bootstrapping – based on 1000 repetitions. Reference categories: MSA; below Federal Poverty Line (FPL).

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

Table 3

Supply adjusted for nearby children (two-stage floating catchment area method) in Minnesota, 2015, by race and ethnicity category.

	White	African-American	Hispanic	American Indian	Asian	Other
Slots adjusted for nearby children						
Any provider	0.564	0.541***	0.562	0.636*	0.496***	0.534***
Child care centers (CCC)	0.210	0.323***	0.256***	0.155***	0.283***	0.260***
Family child care (FCC)	0.296	0.171***	0.247***	0.256*	0.172***	0.218***
Public providers	0.058	0.047***	0.058	0.224***	0.041***	0.056
Share of (adjusted) slots						
Child care centers (CCC)	0.372	0.597***	0.456***	0.244***	0.571***	0.487***
Family child care (FCC)	0.525	0.316***	0.440***	0.403*	0.347***	0.408***
Public providers	0.103	0.087***	0.103	0.352***	0.083***	0.105

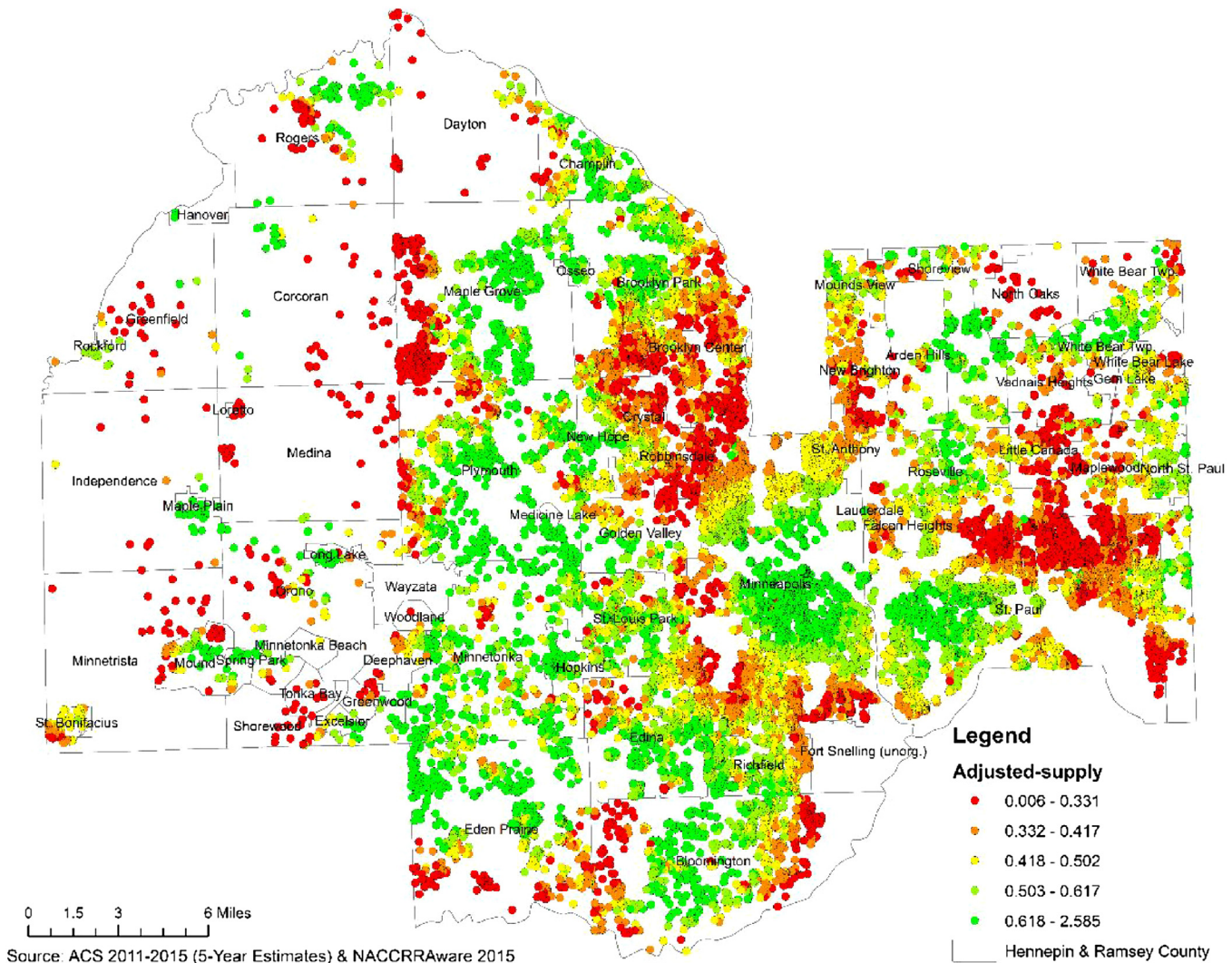
Note: The confidence intervals are computed using bootstrapping – based on 1000 repetitions. Reference category: White.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

locations and families' adjusted supply values can be plotted on a map to illuminate spatial patterns of access to adjusted supply.

Adjusted supply varies considerably around the state, but images of the statewide map are difficult to see in print. Instead, we illustrate the spatial pattern in two counties, Hennepin and Ramsey, which contain the cities of Minneapolis and St. Paul, respectively, along with some nearby suburbs. In Fig. 1, each dot on the map represents the location of a synthetic family. Green dots indicate family

locations with higher access to adjusted slots and red dots indicate lower access. Each color on the map contains an equal number of synthetic families grouped into five quintiles of the adjusted supply measure. Recall that adjusted supply is based on the driving time area around the family's location and is not restricted by census-tract boundaries, although these boundaries are shown on the map for reference. The density of the dot clusters represents the residential density of families with young children. Dense clusters of

**Fig. 1.** Spatial distribution of families' access to supply adjusted for local child population in Hennepin and Ramsey Counties, 2015.

Note: The adjusted supply measure is calculated using the E2SFCA method with Gaussian weight $\beta = 4$ assuming a 20-min driving time catchment area around each family location and each ECE provider.

Table 4

Total access costs and its components based on the closest ECE provider, Minnesota, 2015, by region.

Mean	Statewide	Minneapolis – St. Paul metro area	Greater Minnesota
Driving time (minutes)			
Any provider	2.05	1.28	3.28***
Child care centers (CCC)	7.09	2.97	13.66***
Family child care (FCC)	2.27	1.47	3.54***
Public providers	5.58	4.51	7.30***
Weekly price (dollars per week) at closest provider			
Any provider	147.5	172.5	107.7***
Child care centers (CCC)	210.3	246.2	153.0***
Family child care (FCC)	144.2	162.9	114.3***
Public providers	0	0	0
Total access costs (dollars per week) at closest provider			
Any provider	151.0	174.6	113.2***
Child care centers (CCC)	222.1	251.1	175.8***
Family child care (FCC)	147.9	165.3	120.2***
Public providers	9.3	7.5	12.2***

Note: The confidence intervals are computed using bootstrapping – based on 1000 repetitions. Weekly price is based on a weighted average of prices for each age group. Reference categories: MSA; below Federal Poverty Line (FPL).

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

red indicate areas where both (1) many young children live and (2) the families have relatively little adjusted supply nearby. Seeing the spatial pattern of families' locations with their adjusted supply allows planners, policymakers, and entrepreneurs to identify areas where investments in additional capacity may be desirable.

4.2. Total access cost

Though measures of supply provide important information about access to ECE services, access also depends on cost to the family. We first summarize the cost of access in the local market for each family by calculating the cost to access the provider closest to the family's location based on driving time. Using this approach, we find that on average in Minnesota there is a provider within about two minutes driving time of families with young children, usually a family child care provider (Table 4). Travel time to the closest child care center is higher than for the other types of providers, averaging over seven minutes statewide. Not surprisingly, families in Greater Minnesota, which is more rural, face longer average driving times to their closest provider than do families in the Minneapolis-St. Paul metro area. Driving time to the closest ECE provider for metro families averages just over one minute compared to over three minutes in Greater Minnesota.

At the closest provider to the family, the average weekly price follows expected patterns. The average price is higher for families' closest child care center than for their closest family child care provider, and average prices are much higher for families in the Minneapolis-St. Paul metro area than for Greater Minnesota families. Combining the travel costs based on driving time and weekly price, the total cost to access each family's single closest provider averages \$151 statewide. Not surprisingly, the total costs vary considerably across the provider types. To access the closest child care center, families statewide would incur an average cost of nearly \$222 compared to the average cost of \$148 at families' closest family child care provider. Public providers are assumed to have zero money prices for families and, therefore, access costs are based only on travel time.

The measures based on the closest provider to the family location provide useful summary information about the ECE market near the family, but do not account for differences in capacity or child population nearby nor do they represent families' view of the larger ECE market beyond their single closest provider. Using the adjusted supply measure described in the previous section as a

provider weight, we re-examine the total cost (and its components) using an average across providers nearby each family's location. Closer providers, those with more capacity, and those with fewer children nearer by them are weighted more heavily than others. Results are summarized by region and by income in Table 5.

Including additional providers mechanically raises the average driving time compared to the closest-provider measure but the spatial patterns are similar across the two methods. Families are farther from child care centers, especially in Greater Minnesota, with average weighted driving times nearly four minutes longer than to family child care providers. Weekly prices are slightly higher when weighted by adjusted supply compared to the average price of the closest provider, but the patterns across types of care and regions are the same. The total cost to Minnesota families averaged \$172 per week statewide for all types of care, with higher costs for accessing centers than family child care (or public providers). For families in the Minneapolis-St Paul metro area, the average total cost is \$261 for center-based care compared to \$170 for family child care.

Families with low incomes tend to have slightly shorter average driving time compared to middle and higher income families, and slightly but significantly lower total cost regardless of provider type (Table 5). The total (weighted) cost to access a child care center is \$220 on average for a low-income family compared to \$232 for families with incomes above 185% FPL. Note that in general, the differences in the measures by income group are smaller than the differences by region.

Comparing the total costs of access across race and ethnicity subgroups, we see similar patterns across the types of providers, with access costs highest for child care centers and lowest for public providers (Table 6). Similar cost patterns are seen for the race and ethnicity categories whether using only the closest provider or the adjusted-supply provider weights. The total costs to access center or family child care are lower for American Indian families than other groups because of lower prices nearby. In contrast, longer drive times result in higher travel costs to public providers for American Indian families. To access nearby child care centers, white, non-Hispanic families would need an average of \$224 per week compared to \$258 for African-American families and \$265 for Asian families. Average costs to access centers are similar for Hispanic families (\$227) to white non-Hispanic families (\$224), and lower for American Indian families (\$204). Because ECE prices are higher in the Minneapolis-St. Paul metro area, groups whose pop-

Table 5

Total access costs and its components based on the E2SFCA-weighted providers within 20-min driving time, Minnesota, 2015, by region, by income.

	By region			By income group		
	Statewide	Minneapolis – St Paul metro area	Greater Minnesota	Below FPL	Above FPL and less than 185% FPL	Above 185% FPL
Driving time (minutes)						
Any provider	4.58	4.23	5.13 ^{***}	4.3	4.6 ^{***}	4.6 ^{***}
Child care centers (CCC)	8.53	4.87	14.37 ^{***}	9.1	9.1	8.3 ^{***}
Family child care (FCC)	4.63	4.19	5.34 ^{***}	4.5	4.7 ^{***}	4.6 ^{***}
Public providers	6.32	5.43	7.74 ^{***}	5.4	6.2 ^{***}	6.5 ^{***}
Weekly price (dollars per week)						
Any provider	164.6	200.4	107.3 ^{***}	154.2	157.4 ^{***}	167.7 ^{***}
Child care centers (CCC)	214.8	252.9	154.1 ^{***}	205.2	207.8 ^{***}	217.8 ^{***}
Family child care (FCC)	144.3	163.0	114.5 ^{***}	147.3	148.8 ^{***}	153.5 ^{***}
Public providers	0.0	0.0	0.0	0.0	0.0	0.0
Total access costs (dollars per week)						
Any provider	172.2	207.5	115.9 ^{***}	161.4	165.1 ^{***}	175.4 ^{***}
Child care centers (CCC)	229.1	261.0	178.1 ^{***}	220.3	223.0 ^{***}	231.7 ^{***}
Family child care (FCC)	152.0	170.0	123.4 ^{***}	147.3	148.8 ^{***}	153.5 ^{***}
Public providers	10.5	9.1	12.9 ^{***}	9.0	10.3 ^{***}	10.9 ^{***}

Note: The confidence intervals are computed using bootstrapping – based on 1000 repetitions. Weekly price is based on a weighted average of prices for each age group. Reference categories: MSA; below Federal Poverty Line (FPL).

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

Table 6

Total access costs and its components based on the E2SFCA weighted providers within 20-min driving time, Minnesota, 2015, by race and ethnicity category.

Weighted mean	White	African-American	Hispanic	American Indian	Asian
Driving time (minutes)					
Any provider	4.7	4.1 ^{***}	4.0 ^{***}	5.7 ^{***}	4.2 ^{***}
Child care centers (CCC)	9.2	4.5 ^{***}	5.8 ^{***}	26.4 ^{***}	4.9 ^{***}
Family child care (FCC)	4.7	4.1 ^{***}	4.0 ^{***}	9.8 ^{***}	4.2 ^{***}
Public providers	6.8	4.5 ^{***}	4.6 ^{***}	7.7 ^{***}	5.1 ^{***}
Weekly price (dollars per week)					
Any provider	158.5	198.7 ^{***}	169.6 ^{***}	86.8 ^{***}	201.3 ^{***}
Child care centers (CCC)	208.1	250.1 ^{***}	217.6 ^{***}	160.2 ^{***}	256.5 ^{***}
Family child care (FCC)	141.3	160.6 ^{***}	146.6 ^{***}	120.1 ^{***}	161.9 ^{***}
Public providers	0.0	0.0	0.0	0.0	0.0
Total access costs (dollars per week)					
Any provider	166.4	205.6 ^{***}	176.2 ^{***}	96.3 ^{***}	208.3 ^{***}
Child care centers (CCC)	223.5	257.6 ^{***}	227.3 ^{***}	204.2 ^{***}	264.7 ^{***}
Family child care (FCC)	149.1	167.6 ^{***}	153.2 ^{***}	136.3 ^{***}	168.8 ^{***}
Public providers	11.3	7.5 ^{***}	7.7 ^{***}	12.8 ^{***}	8.5 ^{***}

Note: The confidence intervals are computed using bootstrapping – based on 1000 repetitions. Weekly price is based on a weighted average of prices for each age group. Reference category: White.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

ulation is more heavily concentrated in the metro area face higher prices and higher total access costs.

4.3. Access to quality

We next use these measures to evaluate access to highly rated providers across regions and socio-demographic groups. Providers are categorized in three groups: highly-rated private providers (those with a three- or four-star rating in Parent Aware who report prices), public providers (which are Head Start, Early Head Start or school-based prekindergarten programs), and all other providers (including child care centers and licensed family providers who have no rating or a one- or two-star rating).⁶ Results for the quan-

tity and cost measures for these three groups are shown in Table 7. Families in the Minneapolis-St. Paul metro area have access to an average of 0.13 adjusted slots at highly-rated private providers. In other words, there are about eight nearby young children for every highly-rated, private slot. Families in Greater Minnesota have access to far less highly-rated market supply, with an average of only 0.05 highly rated private adjusted slots nearby. The opposite geographic pattern is seen for public providers, who are also highly rated in the Parent Aware QRIS. Families in Greater Minnesota average 0.10 adjusted slots of public supply while families in the Minneapolis-St Paul metro area average only 0.03. Combining both market and public sources of highly-rated supply, the share of adjusted slots that are highly rated is slightly higher for families

⁶ ECE providers who are not rated in the Parent Aware QRIS may be high quality but have not volunteered to participate in the rating system. Note, however, that the unrated providers and lower-rated providers have similar prices on average. We combine the unrated and lower-rated providers in order to distinguish them from the providers who are highly rated in the QRIS. We keep public providers separate

from other highly-rated providers because of the poor quality of data we have on capacity, enrollment, and price for the public providers, resulting in less confidence in the results for public providers.

Table 7
Adjusted supply and cost measures by quality rating, by region, by income.

	By region			By income		
	Statewide	Minneapolis – St Paul metro area	Greater Minnesota	Below FPL	Above FPL and less than 185% FPL	Above 185% FPL
Slots adjusted for nearby children						
Highly-rated market providers	0.099	0.128	0.054***	0.100	0.097***	0.099
Public providers (highly rated)	0.058	0.031	0.100***	0.081	0.066***	0.051***
All other providers	0.401	0.350	0.482***	0.408	0.398***	0.400***
Share of (adjusted) slots						
Highly-rated market providers	0.177	0.251	0.085***	0.170	0.173***	0.180
Public providers (highly rated)	0.104	0.061	0.157***	0.138	0.118***	0.093***
All other providers	0.719	0.688	0.758***	0.693	0.709***	0.727***
Driving time (minutes)						
Highly-rated market providers	10.5	5.8	18.1***	11.1	11.2	10.3***
Public providers (highly rated)	6.3	5.4	7.7***	5.4	6.2***	6.5***
All other providers	4.6	4.2	5.3***	4.5	4.8***	4.7***
Weekly price (dollars per week)						
Highly-rated market providers	216.7	264.5	140.4***	207.5	209.8***	219.6***
Public providers (highly rated)	0	0	0	0	0	0
All other providers	166.5	195.3	120.6***	161.7	162.1	168.1***
Total access costs (dollars per week)						
Highly-rated market providers	234.3	274.2	170.6***	226.0	228.4***	236.8***
Public providers (highly rated)	10.5	9.1	12.9***	9.0	10.3***	10.9***
All other providers	174.3	202.4	129.4***	169.3	170.0***	175.9***
Quality premium	1.359	1.372	1.344***	1.343	1.355***	1.363***

Note: Highly rated market providers are those with a three- or four-star rating and who charge a non-zero price to parents. Public programs include Head Start, Early Head Start and School-Readiness pre-kindergarten programs, which are assigned four-star ratings in Minnesota's quality rating system. All other providers include private (market) providers who either do not participate in the rating system or have a one- or two-star rating. Weekly price is based on a weighted average of prices for each age group. The quality premium is defined as the ratio of average (weekly) total cost (travel costs plus price) at highly-rated providers to all other (non-public) providers within the catchment area. Measures in this table are based on the enhanced 2SFCA method using a 20-min catchment area and Gaussian distance weight function with beta equal to four. The confidence intervals are computed using bootstrapping – based on 1000 repetitions. Reference categories: MSA; below Federal Poverty Line (FPL).

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

in the Greater Minnesota (31% versus 24%) and a larger proportion of those are public providers.

Travel times to highly-rated private providers are much longer, on average, for Greater Minnesota families than metro families (18 vs. 6 min), but prices are considerably lower. Therefore, total access costs for highly rated private providers are considerably higher in the Minneapolis-St. Paul metro area relative to Greater Minnesota. The weighted average cost to access a highly rated private provider in the metro area is \$274 per week compared to \$171 in Greater Minnesota. Note that we can use the total access cost measure to calculate the “quality premium,” that is, the cost to access a highly rated private provider (including both travel costs and weekly price) compared to other private providers. On average across the state the quality premium was about 1.36, or 36%, and was slightly higher in the Minneapolis-St Paul region (37%) compared to the rest of Minnesota (34%). Families would pay, on average, 36% more to access a highly rated private provider compared to a lower rated or unrated private provider.

Access to highly rated providers varies across income groups, although the differences are smaller than across regions. Families with incomes below the poverty line tend to have higher access to both highly-rated market providers and public providers. The quality premium is smaller for low-income families as well, 34% compared to 36% for middle and higher-income families.

4.4. Comparison of area-based and distance-based measures

The new, distance-based measures account for the fact that families can cross arbitrary administrative boundaries, relaxing area-based measures' implicit assumption of impenetrable boundaries and better reflecting the real experiences of families and providers. Although area-based and distance-based measures could theoretically produce similar findings with regard to the spatial distribution of access, this is not the case in Minnesota. To

demonstrate the difference, we compute both a standard area-based measure and our distance-based supply measure using the same data in the same geographic area (Hennepin and Ramsey Counties) and display the two results in Figs. 2 and 3, respectively. In Fig. 2, the number of slots per young child is calculated for each census tract, a standard area-based measure. In Fig. 3, adjusted supply is calculated for each synthetic family in the census tract, accounting for providers within each family's drive-time proximity and the number of other children near that supply, whether or not the reference family, the providers, or the other families share a census tract. This adjusted supply measure is then averaged across families within each census tract to get a tract-level measure comparable to Fig. 2's area-based measure. The colors in each figure represent quintiles of the distribution of the relevant measure.

By ignoring arbitrary administrative boundaries, the distance-based measure includes less noise than the area-based measure. Comparing the two maps in Figs. 2 and 3, there are two important features demonstrating this difference. First, the distanced-based method generates smoother multi-tract areas of high or low supply because families are allowed to cross boundaries to get to nearby providers. The presence of a large provider in one tract is recognized as adding supply for nearby families in neighboring tracts. In contrast, with the area-based measure, the map shows starkly different values across neighboring areas as it spikes the numerator in one area but contributes nothing to access of families' in neighboring areas.⁷ Second, the scales are different on the two maps, with the area-based measure having more extreme values. There is at least one census tract with no child care providers and, therefore, no slots within its boundaries (slots per child = 0) but in the distance-based approach, the minimum value is 0.14 adjusted slots per child. Similarly, at the high end of the range, one census tract has more

⁷ This is also illustrated in our example in the online supplemental material.

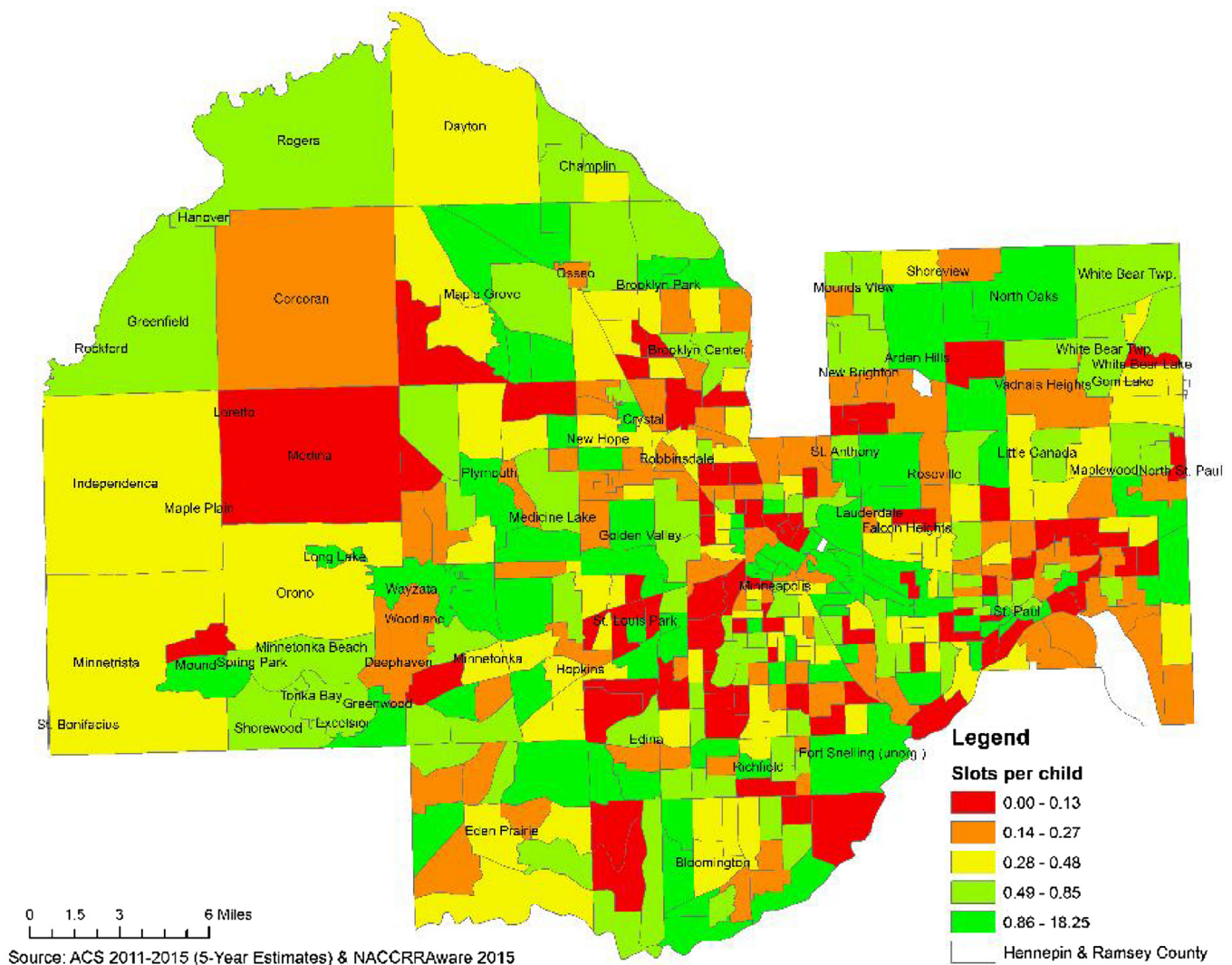


Fig. 2. Slots per child by census tract (area-based measure) in Hennepin and Ramsey Counties, 2015.

than 18 slots per child, driven by a center located in a tract without many children. Yet families will come from nearby census tracts to that center. The distance-based measure of slots adjusted for nearby child population results in a different, more accurate map of areas with low and high relative supply. Whether area-based and distance-based measures will always show different results depends on the size of the geographic units used as the basis of the area-based measure and how close families and providers are located to the boundaries.

How do the three primary dimensions of families' access to ECE services – quantity, cost, and quality – relate to one another and how do these relationships differ when using distance-based measures versus area-based measures? Many underlying economic, demographic, or policy forces could induce correlation between the dimensions. For instance, areas with a greater quantity of care accessible might tend to have lower costs because of greater competition, or instead, might have higher costs because greater underlying effective demand in that community drew in extra capacity and raised prices. On the other hand, if no correlations are observed this could be either because the dimensions truly are uncorrelated or because noisy measurement attenuates the correlation.

We examine the correlations for both distance- and area-based measures in Table 8. The analysis is done on variables averaged up

to the Census tract level to ensure comparability between measure types. Overall the correlations are small, indicating that the cost and quantity measures are not closely related. Cost has a -0.11 correlation with quantity using distance-based measures (adjusted supply with total cost) and a -0.06 correlation with area-based measures (slots-per-tot with weekly price). The correlation based on weekly price is similar between the two measures (-0.08 vs. -0.06) but the distance-based measure also incorporates the fact that less supply nearer by tends to raise travel costs (correlation between adjusted supply and travel cost is -0.35), and so raise the total cost of access. Highly-rated share has a 0.08 correlation with quantity under distance-based measures and a -0.02 correlation under area-based measures. While these correlations are small, the correlations among the distance-based measures are generally significantly larger (in absolute value) than those of the area-based measures. Very small correlations for the zone-based measures suggest they are noisy measures and not closely related to each other.

5. Discussion and conclusion

This study, using GIS-based simulation methods, introduces new measures of access that are family-centered and reflect the quantity, cost and quality dimensions of access to early care and

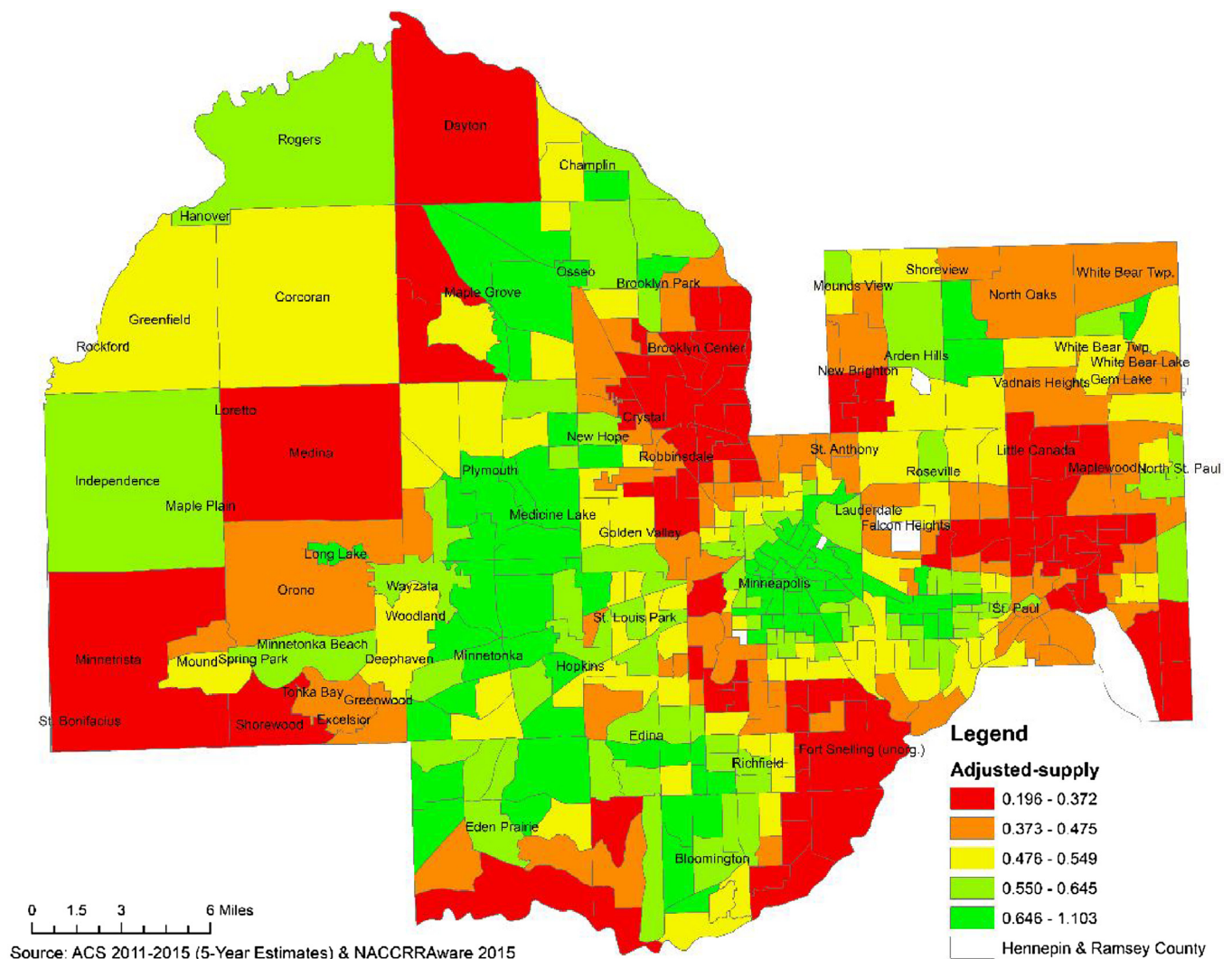


Fig. 3. E2SFCA adjusted supply measure (distance-based measure) averaged by census tract, in Hennepin and Ramsey Counties, 2015.

Table 8

Correlations of quantity measures with cost and quality measures by measure type and region.

	Statewide		Minneapolis – St Paul metro area		Greater Minnesota	
	Distance-based (Adj-supply)	Area-based (Slots per child)	Distance-based (Adj-supply)	Area-based (Slots per child)	Distance-based (Adj-supply)	Area-based (Slots per child)
Cost measures						
Weekly Price	−0.08***	−0.06	0.34***	−0.12	0.02***	0.04
Travel Cost	−0.35***	–	−0.38***	–	−0.46***	–
Total Cost	−0.11***	–	0.33***	–	−0.09**	–
Quality measures						
Share Highly-rated	0.08**	−0.02	0.02	0.02	0.23***	−0.05
Share Public	0.18***	0.03	−0.22***	0.00	0.21***	0.03
Share Highly-rated, market	−0.05	−0.04	0.13***	0.02	0.06*	−0.10**

Note: Unit of analysis is census-tracts (i.e., distance-based family-level measures are aggregated (averaged) up to census-tract for purpose of comparison); The distance-based measures are adjusted supply and adjusted-supply weighted measures of cost and quality shares, as introduced in this paper. The zone-based measures are conventional slots-per-child, slot-weighted average weekly price and share of slots highly rated by type. The confidence intervals are computed using bootstrapping – based on 1000 repetitions:

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

education services. The literature has acknowledged that the lack of data sources is the biggest impediment to the study of community-level child care accessibility (Gordon & Chase-Lansdale, 2001). Measures such as number of slots per child per county or num-

ber of child care workers per child within a 20–30 mile radius of the child's ZIP code of residence have been suggested as measures of child care availability across different communities in U.S. (Coley, Votruba-Drzal, Collins, & Miller, 2014; Gordon & Chase-

Lansdale, 2001). Our study builds upon the previous literature on child care access measures by developing a GIS-based access measure that utilizes a distance-based approach. Using licensed child care provider data with detailed provider characteristics and simulated family locations (based on population and race data from the Census), we calculate travel time between families and neighboring providers to examine access disparities across locations and demographic groups in Minnesota.

Conceptually, the distance-based approach is intended to represent the family's view of the ECE market rather than the family's ability to access a particular slot. By using a family-centered approach, the measures improve upon existing area-based methods of measuring supply. Measures based on the number or capacity of providers within an area such as a census tract or ZIP code area provide insufficient information about the availability of providers close to families who live nearby but on the other side of the administrative boundaries. In some studies, area-based measures are dichotomized to identify areas with low supply or shortages (e.g., "child care deserts") versus those with adequate supply (Dobbins, Tercha, McCreedy, & Liu, 2016; Malik, Hamm, Adamu, & Morrissey, 2016). These approaches have the advantage of being relatively straightforward, transparent and emotionally impactful. No one wants children to be stuck in a desert. These designations may be misleading, however, because of the restriction of the area boundaries and the loss of information in converting from a continuous slots-per-child measure to a dichotomous desert-or-not measure. Another attraction, though not virtue, of the dichotomous measure is that it masks the underlying area-based measure's noisiness. In contrast, our new measures allow for crossing these boundaries and provide a smooth, continuous measure of supply adjusted for nearby population. Thus, while our adjusted supply measure is conceptually similar to an area-based measure of the number of slots per child in a given area, the view of variation in access may be more accurate as well as more nuanced using the new measure. While the new measures could be used to define a threshold for a child care desert, the continuous measure may provide more useful information for policymakers and planners.

These measures of supply provide important information about the availability of ECE providers nearby to families but a family's ability to access a particular provider also depends upon the cost to the family. Our total cost of access measure captures both the price the provider charges and the proximity to the family in dollar terms in a single measure. We demonstrate that total access costs to the family can be represented either by the cost to access the closest provider, or by using a weighted average of providers within a driving time catchment area around the family's location. The results show similar patterns using either approach. In the extant literature, a common area-based measure of cost is the average price of providers in an area, which ignores the spatial distribution of families, and therefore, gives a limited picture of the prices of providers nearby the family. Average price alone also ignores travel time costs.

Given a measure of quality of ECE services, similar quantity and cost measures can be estimated for providers of differing quality levels. While Minnesota's QRIS is relatively new and has a limited coverage, in 2015 the differences in the adjusted quantity and cost of accessing highly rated providers in different regions of the state were large. The quality premium, as measured by the ratio of total access costs at a highly rated market provider relative to an unrated or lower-rated provider, provides a summary measure of the tradeoff a family faces in terms of cost and quality. This measure incorporates both travel costs and provider prices, and so measures how much more it would cost a family to access highly rated market-based care. Statewide, the quality premium is estimated to be 36%, indicating that families would incur costs 36% higher to access a highly-rated market provider compared to an unrated

or lower-rated provider. The average premium estimated in this study is quite similar to the cost difference Mocan (1997) calculates (38%) for providing 'good' quality child care services compared to "mediocre" quality (his cost difference was \$243 to \$324 in 1993 dollars). The quality premium varies considerably across families, depending on how close they live to a highly rated provider, and the price charged.

The findings of this study suggest that conclusions one can draw about child care accessibility in different communities can sharply differ depending on how access measures are calculated. By using readily available ACS data and GIS tools to estimate household-to-provider distance-based measures, this study avoids the strict assumptions of area-based measures. The improved measures of accessibility yield better understanding of unmet childcare needs of different communities across the state.

5.1. Study limitations

While this study provides new distance-based measures of access to high quality ECE, there are a number of limitations and directions for future research. An important limitation is that we do not have data on precise residential locations of families who are seeking or using child care; therefore, there is the measurement error in the locations and travel times. In addition, we are assuming travel from home to child care by car and are not able to incorporate travel between work location and child care or by other modes of transportation in the analysis. We do not have direct observational measures of quality, rather we rely on the state's quality rating system as an indicator of quality. Further, the low level of participation in Parent Aware among providers means that the "unrated" category includes providers of varying quality levels. As the Parent Aware QRIS system matures and more providers join, measured access to highly rated ECE will change. In addition, because of data limitations, the current study does not account for how public, voucher-based subsidies affect the cost of child care for some families. As better data become available on the capacity and site locations for public providers, we plan to expand the analysis of how public investments support access to high quality ECE.

The access measures derived in this paper are measures of the supply of child care and early learning opportunities. While these distance-based measures are a substantial improvement over current measures, they do not account for the availability of openings at the nearest provider, nor account for family preferences with regards to care provided by relatives, or specific language or cultural preferences. Despite these limitations, the access measures provide new insights into the access and affordability of early care and learning opportunities for families of different income levels and sociodemographic groups.

5.2. Conclusion

The concept of access is complex, and there is no commonly accepted definition or measure of access in the literature on early care and education. This paper expands our understanding of access by combining information on the locations of families and providers with data on prices to account for both costs and availability of supply. By developing new access measures that are centered on the family's location rather than an area-based aggregate measure, the results provide more accurate information on the communities in which families have better or worse access to early care and education. The new measures incorporate both cost and proximity to examine the options available to families. As the Parent Aware quality rating system expands in Minnesota, changes in access to high quality early care and education can be tracked. The spatial distributions of highly rated providers and families interact, resulting in differential levels of access to quality ECE across geographic space.

Families may live close to high quality providers, yet be unable to afford them. Information from these access measures can help give a rich multi-dimensional picture of the local market and be used by policymakers to determine where to support expansions of supply and other mechanisms such as subsidies and scholarships to improve access to high quality early care and education for all children.

Acknowledgements

This work was supported by the University of Minnesota Faculty Grant-in-Aid program and the Center for Urban and Regional Affairs (CURA) Faculty Interactive Research Program. Davis also acknowledges the support of the USDA National Institute of Food and Agriculture, Hatch Multistate Project NE-1049 and Minnesota Agricultural Experiment Station Project MIN-14-081. The funding sources had no involvement in the study design, collection, analysis and interpretation of the data, writing of the report, or the decision to submit the article for publication.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ecresq.2018.08.001>.

References

- Alexander, D., Breslay, L., Claessens, A., Greenberg, E., Henly, J., Sandstrom, H., & Stoll, M. (2016). *Child Care Market Conditions and Low-Income Families' Needs: Identifying Mismatches in Four Select Communities. Presented at the APPAM Fall Research Conference Washington D.C.*
- Bassok, D., Fitzpatrick, M., & Loeb, S. (2011). *Disparities in child care availability across communities: Differential reflection of targeted interventions and local demand. CEPA Working Paper.* Retrieved from: <http://cepa.stanford.edu/content/disparities-child-care-availability-across-communities-differential-reflection-targeted>
- Bassok, D., & Galdo, E. (2016). Inequality in preschool quality? Community-level disparities in access to high-quality learning environments. *Early Education and Development*, 27(1), 128–144. <http://dx.doi.org/10.1080/10409289.2015.1057463>
- Carrillo, D., Harknett, K., Logan, A., Luhr, S., & Schneider, D. (2017). Instability of work and care: How work schedules shape child-care arrangements for parents working in the service sector. *Social Service Review*, 91(3), 422–455. <http://dx.doi.org/10.1086/693750>
- Casper, L. M., & O'Connell, M. (1998). Work, income, the economy, and married fathers as child-care providers. *Demography*, 35(2), 243–250. <http://dx.doi.org/10.2307/3004055>
- Cassidy, D. J., Hestenes, L. L., Hansen, J. K., Hegde, A., Shim, J., & Hestenes, S. (2005). Revisiting the two faces of child care quality: Structure and process. *Early Education and Development*, 16(4), 505–520. <http://dx.doi.org/10.1207/s15566935eed1604.10>
- Chipty, T., & Witte, A. D. (1998). *Effects of Information Provision in a Vertically Differentiated Market (Working Paper No. 6493).* National Bureau of Economic Research. Retrieved from: <http://www.nber.org/papers/w6493>
- Cleveland, J., Li, W., Orfali, N., & Tout, K. (2016). *Statewide expansion of Parent Aware ratings: Year 4 report.* Minneapolis, MN: Child Trends.
- Cochi Ficano, C. K. (2006). Child-care market mechanisms: Does policy affect the quantity of care. *Social Service Review*, 80(3), 453–484. <http://dx.doi.org/10.1086/505447>
- Coley, R. L., Votruba-Drzal, E., Collins, M. A., & Miller, P. (2014). Selection into early education and care settings: Differences by developmental period. *Early Childhood Research Quarterly*, 29(3), 319–332. <http://dx.doi.org/10.1016/j.ecresq.2014.03.006>
- Covington, K. (2007). Evidence of dynamic geographic shifts in metropolitan child care markets over the 1990. *International Journal of Economic Development*, 9(3), 159.
- Cryer, D., Tietze, W., & Wessels, H. (2002). Parents' perceptions of their children's child care: A cross-national comparison. *Early Childhood Research Quarterly*, 17(2), 259–277. [http://dx.doi.org/10.1016/S0885-2006\(02\)00148-5](http://dx.doi.org/10.1016/S0885-2006(02)00148-5)
- Davis, E. E., & Connelly, R. (2005). The influence of local price and availability on parents' choice of child care. *Population Research and Policy Review*, 24(4), 301–334. <http://dx.doi.org/10.1007/s11113-005-8515-y>
- Dobbins, D., Tercha, J., McCreedy, M., & Liu, A. (2016). *Child Care Deserts.* Retrieved from: <http://usa.childcareaware.org/advocacy-public-policy/resources/research/childcaredeserts2016/>
- Fischer, R. L., Nelson, L., Mikelbank, K., & Coulton, C. (2008). Space to learn and grow: Assessing the capacity of a regional early care and education system. *Child & Youth Care Forum*, 37(2), 75–86. <http://dx.doi.org/10.1007/s10566-008-9049-3>
- Friesen, S., Lin, V.-K., Forry, N., & Tout, K. (2017). *Defining and Measuring Access to High Quality Early Care and Education: A Guidebook for Policymakers and Researchers (No. OPRE Report #2017-08).* Office of Planning, Research and Evaluation, Administration for Children and Families U.S. Department of Health and Human Services.
- Fuller, B., & Liang, X. (1996). Market failure? Estimating inequality in preschool availability. *Educational Evaluation and Policy Analysis*, 18(1), 31–49.
- Gehlke, C. E., & Biehl, K. (1934). Certain effects of grouping upon the size of the correlation coefficient in census tract material. *Journal of the American Statistical Association*, 29(185A), 169–170. <http://dx.doi.org/10.1080/01621459.1934.10506247>
- Gordon, R., & Chase-Lansdale, L. (2001). Availability of child care in the United States: A description and analysis of data sources. *Demography*, 38(2), 299–316.
- Han, W.-J. (2004). Nonstandard work schedules and child care decisions: Evidence from the NICHD Study of Early Child Care. *Early Childhood Research Quarterly*, 19(2), 231–256. <http://dx.doi.org/10.1016/j.ecresq.2004.04.003>
- Hardy, E., Ha, Y., Crisan, U., Giapponi, K., Joshi, P., & Geronimo, K. (2016). Mapping Local Child Care Subsidy Delivery Systems in Massachusetts. Presented at the APPAM Fall Research Conference Washington D.C.
- Hatfield, B. E., Lower, J. K., Cassidy, D. J., & Faldowski, R. A. (2015). Inequities in access to quality early care and education: Associations with funding and community context. *Early Childhood Research Quarterly*, 30(Part B), 316–326. <http://dx.doi.org/10.1016/j.ecresq.2014.01.001>
- Hofferth, S. L., & Wissoker, D. A. (1992). Price, quality, and income in child care choice. *The Journal of Human Resources*, 27(1), 70–111. <http://dx.doi.org/10.2307/145913>
- Huber, S., & Rust, C. (2016). Calculate travel time and distance with OpenStreetMap data using the Open Source Routing Machine (OSRM). *Stata Journal*, 16(2), 416–423.
- Jacobson, L. (2001). *Understanding Child Care Demand and Supply Issues: New Lessons from Los Angeles.* PACE Policy Brief. Early Education Series. Retrieved from: <https://eric.ed.gov/?id=ED455029>
- Johnson, A. D., Ryan, R. M., & Brooks-Gunn, J. (2012). Child-care subsidies: Do they impact the quality of care children experience. *Child Development*, 83(4), 1444–1461. <http://dx.doi.org/10.1111/j.1467-8624.2012.1780.x>
- Kim, J., & Fram, M. S. (2009). Profiles of choice: Parents' patterns of priority in child care decision-making. *Early Childhood Research Quarterly*, 24(1), 77–91. <http://dx.doi.org/10.1016/j.ecresq.2008.10.001>
- Liang, X., Fuller, B., & Singer, J. D. (2000). Ethnic differences in child care selection: the influence of family structure, parental practices, and home language. *Early Childhood Research Quarterly*, 15(3), 357–384. [http://dx.doi.org/10.1016/S0885-2006\(00\)00071-5](http://dx.doi.org/10.1016/S0885-2006(00)00071-5)
- Loeb, S., Fuller, B., Kagan, S. L., & Carrol, B. (2004). Child care in poor communities: Early learning effects of type, quality, and stability. *Child Development*, 75(1), 47–65. <http://dx.doi.org/10.1111/j.1467-8624.2004.00653.x>
- Luo, W., & Qi, Y. (2009). An enhanced two-step floating catchment area (E2SFCA) method for measuring spatial accessibility to primary care physicians. *Health & Place*, 15(4), 1100–1107. <http://dx.doi.org/10.1016/j.healthplace.2009.06.002>
- Luo, W., & Wang, F. (2003a). Measures of spatial accessibility to health care in a GIS environment: Synthesis and a case study in the Chicago Region. *Environment and Planning B: Planning and Design*, 30(6), 865–884. <http://dx.doi.org/10.1068/b29120>
- Luo, W., & Wang, F. (2003b). *Spatial Accessibility to Primary Care and Physician Shortage Area Designation: A Case Study in Illinois with GIS Approaches.* pp. 261–279. <http://dx.doi.org/10.4018/978-1-59140-042-4.ch015>
- Malik, R., Hamm, K., Adamu, M., & Morrissey, T. (2016). *Child Care Deserts An Analysis of Child Care Centers by ZIP Code in 8 States.* Center for American Progress. Retrieved from: <https://www.americanprogress.org/issues/early-childhood/reports/2016/10/27/225703/child-care-deserts/>
- Manson, S., Schroeder, J., Van Riper, D., & Ruggles, S. (2017). *IPUMS National Historical Geographic Information System: Version 12.0 [Database].* Minneapolis, MN: University of Minnesota.
- Marshall, N. L., Robeson, W. W., Tracy, A. J., Frye, A., & Roberts, J. (2013). Subsidized child care, maternal employment and access to quality, affordable child care. *Early Childhood Research Quarterly*, 28(4), 808–819. <http://dx.doi.org/10.1016/j.ecresq.2013.07.008>
- McLafferty, S. L. (2003). GIS and health care. *Annual Review of Public Health*, 24(1), 25–42. <http://dx.doi.org/10.1146/annurev.publhealth.24.012902.141012>
- Mocan, H. N. (1997). Cost functions, efficiency, and quality in day care centers. *The Journal of Human Resources*, 32(4), 861–891. <http://dx.doi.org/10.2307/146430>
- National Survey of Early Care and Education Project Team. (2016). *How Far are Early Care and Education Arrangements from Children's Homes? (OPRE Report No. 2016-10).* Retrieved from: <https://www.acf.hhs.gov/opre/resource/how-far-are-early-care-and-education-arrangements-from-childrens-homes>
- Neidell, M., & Waldfogel, J. (2009). Program participation of immigrant children: Evidence from the local availability of Head Start. *Economics of Education Review*, 28(6), 704–715. <http://dx.doi.org/10.1016/j.econedurev.2009.06.004>
- Openshaw, S. (1984). *The modifiable areal unit problem Concepts and Techniques in Modern Geography.* Norwich: Geo Books.
- Overson, K., & McConnell, S. (2015). *Supply and Demand for Early Care and Education Services in North Minneapolis.* Center for Early Education and Development & Urban Research and Outreach/Engagement Center, University

- of Minnesota. Retrieved from. <http://www.cehd.umn.edu/EdPsych/research/supply-and-demand.html>
- Peisner-Feinberg, E. S., Burchinal, M. R., Clifford, R. M., Culkin, M. L., Howes, C., Kagan, S. L., & Yazejian, N. (2001). The relation of preschool child-care quality to children's cognitive and social developmental trajectories through second grade. *Child Development*, 72(5), 1534–1553.
- Peyton, V., Jacobs, A., O'Brien, M., & Roy, C. (2001). Reasons for choosing child care: Associations with family factors, quality, and satisfaction. *Early Childhood Research Quarterly*, 16(2), 191–208. [http://dx.doi.org/10.1016/S0885-2006\(01\)00098-9](http://dx.doi.org/10.1016/S0885-2006(01)00098-9)
- Queralto, M., & Witte, A. D. (1998). Influences on neighborhood supply of child care in Massachusetts. *Social Service Review*, 72(1), 17–46. <http://dx.doi.org/10.1086/515744>
- Radey, M., & Brewster, K. L. (2007). The influence of race/ethnicity on disadvantaged mothers' child care arrangements. *Early Childhood Research Quarterly*, 22(3), 379–393. <http://dx.doi.org/10.1016/j.ecresq.2007.05.004>
- Tout, K., Cleveland, J., Li, W., Starr, R., Soli, M., & Bultinck, E. (2016). *The Parent Aware evaluation: Initial validation report*. Minneapolis, MN: Child Trends.
- US Department of Health & Human Services. (2014). *Child Care and Development Block Grant Act (CCDBG) of 2014: Plain Language Summary of Statutory Changes*. Retrieved from. <https://www.acf.hhs.gov/occ/reso>
- Yamauchi, C. (2010). The availability of child care centers, perceived search costs and parental life satisfaction. *Review of Economics of the Household*, 8(2), 231–253. <http://dx.doi.org/10.1007/s11150-009-9071-8>